Abderrahmane FERRATI



Electrical Engineering and Industrial Control Systems Student

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Final-year internship from mid-January 2027 · 5-6 months or longer · Open geographic mobility

Engineering student at ENSET Mohammedia with an interest in modeling, analysis and control of industrial systems. Seeking a final-year internship in electrical engineering, automation or energy.

PROFESSIONAL EXPERIENCE

YAZAKI Morocco Meknes (YMM1) | June - August 2026

Engineering internship - Individual industrial rooftop solar design study

• Analyzed electricity bills and roof constraints; selected components and ballasted mounting for a **1.20 MWp** study: **1,656 modules and 7 inverters**.

• Sizing calculations and **PVsyst** simulation: **2.18 GWh/year simulated**; simulated demand coverage of **41% in summer and 31% in winter**.

• Created **Revit** BIM models and renders for the rooftop layout. Demonstration dashboard in development: power, energy and simulated faults.

ONCF - Rabat-Agdal | June - August 2025

Introductory internship - Railway systems study and control-law simulation

• Studied signaling equipment and fault detection in track circuits.

• Simulated **P, PI, PD and PID controllers in MATLAB/Simulink** as part of a braking-control study; no operational braking modification is claimed.

EDUCATION

ENSET Mohammedia | 2024 - present - Engineering program in Electrical Engineering and Industrial Control Systems.

Lycée Omar Ibn Al Khattab | 2022 - 2024 - MPSI/MP preparatory classes.

Lycée Tarik | 2022 - Baccalaureate in Mathematical Sciences.

ACADEMIC PROJECTS

Automated filling and capping | Individual project

• Designed a **Siemens S7-300** control system: PO/PC GRAFCET, wiring study and **LADDER programming in STEP 7**.

• Ran **PLCSim and FluidSim** simulations: normal, manual and fault modes, plus a simulated emergency-stop scenario; no safety certification is claimed.

24 V / 15 V Push-Pull converter | Individual project

• Designed a **50 kHz** converter using UC2525 control, a high-frequency transformer, IRF3205 MOSFETs and Schottky diodes.

• **Proteus ISIS** simulation and **ARES** PCB layout. Studied overcurrent, overtemperature and soft-start concepts; no hardware testing is claimed.

Modular educational test bench | Group project

• My contribution: reverse engineering and analysis of the three-phase bench; study of an architecture using **three independent multi-tap 100 VA single-phase transformers** for single-phase and three-phase configurations.

• Mechanical redesign in **SolidWorks** and study of a six-diode rectifier; manufacturing and experimental validation are not established.

TECHNICAL SKILLS

Energy: PVsyst, Autodesk Revit, BIM, self-consumption PV design, techno-economic analysis.

Automation: TIA Portal, STEP 7, GRAFCET, LADDER, FluidSim, S7-300/S7-1500.

Electrical: CANECO BT, AutoCAD Electrical, power/control diagrams, cable sizing.

Embedded systems and programming: Arduino, sensors, data acquisition, HMI, Python, C, Structured Text.

Simulation and design: MATLAB/Simulink, Proteus, SolidWorks.

Reference standards and awareness: IEC 60364, IEC 61131-3, NF C 15-100, ISO 50001; no compliance qualification claimed.

LANGUAGES

Arabic: native · French: fluent · English: C1 (EF SET)

CERTIFICATIONS

Python Essentials 1 & 2 - Cisco Networking Academy · ISO 50001:2018 Awareness · EF SET English Certificate - C1